

A native icosian realization of Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$

The 600-cell unit group reproduces a family of cuspidal Hecke systems — parameter-free and out-of-sample

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Abstract

We give an independent, parameter-free implementation of the standard definite Brandt method for cuspidal Hilbert modular forms over $K = \mathbb{Q}(\sqrt{5})$, driven entirely by the geometry of the 600-cell. The binary icosahedral group $2I$ is the reduced-norm-one unit group of a maximal order $\mathcal{O}$ (the *icosian ring*) in the definite quaternion algebra ramified at the two infinite places of K ; its lattice is E_8 . For a split prime $\mathfrak{p} \subset \mathcal{O}_K$ we build the Brandt module of $\mathcal{O}$ at level $\mathfrak{p}$ purely from the action of $A_5 = 2I/\{\pm 1\}$ on $\mathbb{P}^1(\mathcal{O}_K/\mathfrak{p})$, with no input from elliptic curves, modular symbols, or L -function tables. The resulting Hecke operators are integral and self-adjoint, and we verify against independently point-counted data that this construction reproduces *a family* of forms, not a single example: (i) the complete prime-level cuspidal-dimension sequence over K for norm < 100 (7/7 agreements with Dembél , including the two non-obvious cases — the absent form at norm 59 and the jump to a genus-2 form at norm 61); (ii) the full Hecke eigenvalue sets of *two* rational eigenforms (levels 31 and 41), matched out-of-sample against brute-force point counts of the associated elliptic curves at every prime tested, with no fitting; (iii) the genus-2 form at level 61, whose irrational eigenvalues in $\mathbb{Z}[\varphi]$ (real multiplication by K) the geometry reconstructs, matching the published values exactly at the inert primes. We close with an explicitly heuristic discussion of how this finite, self-adjoint, coefficient-side structure relates to the Riemann Hypothesis, emphasizing the gap rather than bridging it: RH is the *global, analytic* positivity equivalent to the Weil and Li criteria, of which a finite Brandt computation supplies nothing. **This paper proves no Millennium problem and claims no progress on one.** Its contribution is a clean, reproducible geometric realization of known arithmetic, and an honest delimitation of the gap that remains.

Scope

This is a *reproduction-and-map* paper. The arithmetic it recovers is due to Demb   [7] and is an instance of the Jacquet–Langlands correspondence [9, 8]; our contribution is the independent, parameter-free *geometric* route (native icosian arithmetic, no computer-algebra modular package), the explicit no-fit out-of-sample protocol, and a precise statement of what the verified structure does *not* reach (§8). The arithmetic targets are Demb   s known examples; the novelty is independent implementation and provenance, not new forms. We claim *validation and exposition*, not new number theory, and *no* progress on RH itself.

1 Introduction

The 600-cell is the regular convex 4-polytope whose 120 vertices are the unit icosians: the binary icosahedral group $2I \subset \mathbb{H}$ of order 120. Viewed as a lattice in $\mathbb{R}^4 \cong \mathbb{H}$, the icosians are a copy of E_8 over $\mathbb{Z}$, and a copy of the Hurwitz-type maximal order $\mathcal{O}$ in the definite quaternion algebra $B = \left(\frac{-1, -1}{K}\right)$ over $K = \mathbb{Q}(\sqrt{5})$ [5, 13]. This is the smallest arithmetic setting in which the golden ratio $\varphi = (1 + \sqrt{5})/2$ and the icosahedral symmetry that organizes the 600-cell become number theory.

By the Jacquet–Langlands correspondence and the Eichler basis problem [9, 8, 12], the Brandt module of $\mathcal{O}$ at a prime level $\mathfrak{p}$ computes the Hilbert cusp forms of parallel weight 2 and level $\mathfrak{p}$ over K . Demb  le [7] used exactly this to produce the first explicit tables of such forms; the smallest level carrying a cusp form has norm 31, and the associated rational eigenform corresponds to an elliptic curve over K [2].

Our question is narrow and testable: *does the 600-cell geometry, used on its own terms, reproduce this arithmetic?* We answer yes, and quantitatively. The validation is anti-circular by construction: every target we compare against is generated independently — by brute-force point counting of an elliptic curve, or from the published literature — and is used *only in comparison*, never in building the Brandt operators.

Results.

1. **One form, completely (§3).** At level $\mathfrak{p}_{31}$ the geometry gives a 2-dimensional Brandt module, Eichler mass $8/15$, a 1-dimensional cuspidal eigenform, and Hecke eigenvalues matching the point-counted curve 31.1-a1 at all 44 good prime ideals of norm ≤ 200 (over 25 rational-prime norm buckets): at each split norm the two totally positive generators resolve the *complete* eigenvalue pair $\{a_q, a_{q'}\}$, not merely one representative, and the Brandt value at the level ideal itself recovers the Steinberg datum $a_{\mathfrak{p}_{31}} = -1$ used by the Atkin–Lehner check.
2. **A family, by dimension (§4).** The geometric orbit count reproduces the entire prime-level cuspidal-dimension sequence over K for norm < 100 : 7/7 agreement with [7], including the absence of a form at norm 59 and the jump to cuspidal dimension 2 at norm 61.
3. **A family, by eigenvalue (§5).** A second rational eigenform, at level $\mathfrak{p}_{41}$, is built by the same procedure and matched out-of-sample (full eigenvalue sets, 9/9 primes) against the point-counted curve $E_{41} : y^2 + \varphi y = x^3 - \varphi x^2$.
4. **The non-rational form (§6).** At level $\mathfrak{p}_{61}$ the geometry yields cuspidal dimension 2 — the genus-2 abelian surface with real multiplication by K of [7, Rem. 4.4]. From the 3×3 Brandt matrix we recover, at each of the six tested primes (norms 4–49), an eigenvalue whose field-theoretic discriminant is $5 \cdot (\text{square})$, i.e. an irrational element of $\mathbb{Z}[\varphi]$; at the three inert primes the values match the published ones exactly.
5. **The certificate algebra, and a precise non-claim (§7–§8).** The Brandt operators are self-adjoint for the orbit-size measure — an exactly checkable certificate in a small algebra we make precise. A closing section states exactly which analytic statements (Weil/Li positivity, i.e. GRH-type questions for the assembled L -functions) the finite computation does *not* reach, and why a finite coefficient computation cannot reach them.

All computations are pure Python, reproduce in seconds to minutes, and are included with the paper; §9 gives the manifest.

2 The icosian order and its Brandt module

Let $K = \mathbb{Q}(\sqrt{5})$, $\varphi = (1 + \sqrt{5})/2$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$, with $N(a + b\varphi) = a^2 + ab - b^2$ and nontrivial automorphism $\sigma : \varphi \mapsto 1 - \varphi$. Let $B = \left(\frac{-1, -1}{K}\right)$ with standard basis $1, i, j, k$. The *icosian ring* $\mathcal{O}$ is the maximal order whose group of reduced-norm-one units is the binary icosahedral group, $\mathcal{O}^1 = 2I$, $|2I| = 120$; as a $\mathbb{Z}$ -lattice $\mathcal{O}$ is E_8 [5, 13]. B is ramified exactly at the two infinite places of K , hence definite, and (its maximal orders form) a single ideal class.

Throughout we take the level $\mathfrak{p} \subset \mathcal{O}_K$ to be a prime that *splits* in K , so that $N(\mathfrak{p}) = q$ is a rational prime and the residue field is $\mathcal{O}_K/\mathfrak{p} \cong \mathbb{F}_q$. (The levels 31, 41, 61 used below are all split; the inert and ramified cases, with residue field $\mathbb{F}_{q^2}$ or $\mathbb{F}_q$, are not needed.) For odd q fix a local splitting

$$\iota_{\mathfrak{p}} : \mathcal{O} \otimes_{\mathcal{O}_K} (\mathcal{O}_K)_{\mathfrak{p}} \xrightarrow{\sim} M_2(\mathcal{O}_K/\mathfrak{p}) = M_2(\mathbb{F}_q),$$

constructed explicitly by solving $i^2 = j^2 = -1$, $ij = -ji$ over $\mathbb{F}_q$ (possible since -1 is a sum of two squares mod q). The unit group $2I$ acts through $\iota_{\mathfrak{p}}$ on the $q + 1$ points of $\mathbb{P}^1(\mathbb{F}_q)$; since $-1 \in 2I$ acts trivially, the action factors through $A_5 = 2I/\{\pm 1\}$. Because $\mathcal{O}$ has class number one, the right-ideal classes of the Eichler order $\mathcal{O}_0(\mathfrak{p})$ identify with $\mathcal{O}^1 \backslash \mathbb{P}^1(\mathcal{O}_K/\mathfrak{p})$, i.e. with the A_5 -orbits on $\mathbb{P}^1(\mathbb{F}_q)$ (the definite/Brandt method; [13, Ch. 41], [12]). Write h for the number of orbits and $\mu = (\mu_1, \dots, \mu_h)$ for their sizes, so $\sum_i \mu_i = q + 1$.

Brandt operators. For a totally positive $\varpi \in \mathcal{O}_K$ with $N(\varpi) = N(\mathfrak{q})$, $\mathfrak{q} \nmid \mathfrak{p}$, enumerate the icosians of reduced norm ϖ (a short-vector enumeration in E_8). Their number is $120 r(\varpi)$ where $r(\varpi) = N(\mathfrak{q}) + 1$ is the relevant Brandt / theta coefficient for the icosian order at the primes we use; we verify this count at construction time as a gate. Push each element through $\iota_{\mathfrak{p}}$ to a matrix in $\mathrm{GL}_2(\mathbb{F}_q)$, and let it permute $\mathbb{P}^1(\mathbb{F}_q)$. Accumulating the induced map on orbits gives the Brandt matrix $B(\mathfrak{q}) \in M_h(\mathbb{Q})$,

$$B(\mathfrak{q})_{IJ} = \frac{1}{120 \mu_I} \#\{ (x, \gamma) : x \in \text{orbit } I, N(\gamma) = \varpi, x \cdot \iota_{\mathfrak{p}}(\gamma) \in \text{orbit } J \}. \quad (1)$$

These are the Hecke operators $T_{\mathfrak{q}}$ on the Brandt module. Three intrinsic invariants are checked at construction time, before any arithmetic target is consulted:

- *Eisenstein row sums:* $\sum_J B(\mathfrak{q})_{IJ} = N(\mathfrak{q}) + 1$ (the all-ones vector is the Eisenstein eigenvector);
- *integrality:* $B(\mathfrak{q}) \in M_h(\mathbb{Z})$;
- *self-adjointness:* $\mu_I B(\mathfrak{q})_{IJ} = \mu_J B(\mathfrak{q})_{JI}$, i.e. $B(\mathfrak{q})$ is symmetric for the inner product weighted by the orbit sizes μ .

The cuspidal subspace is the μ -orthogonal complement of the Eisenstein vector; on it the eigenvalues of $B(\mathfrak{q})$ are the Hecke eigenvalues $a_{\mathfrak{q}}$ of the cusp forms of level $\mathfrak{p}$.

Scope

Equation (1) and the bijection “classes = A_5 -orbits” are standard for class number one [12, 13]. The point of the paper is that we instantiate them with nothing but icosian short-vector arithmetic and the A_5 -action, and then test the output against independent data.

3 Level 31: one form, completely

At $\mathfrak{p}_{31} = (5\varphi - 2)$, $N(\mathfrak{p}_{31}) = 31$ (the smallest norm carrying a cusp form [7]), the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{31})$ has two orbits, $\mu = (12, 20)$, so $h = 2$ and the cuspidal dimension is 1. The point

stabilizers have orders $60/12 = 5$ and $60/20 = 3$, giving Eichler mass $\frac{1}{5} + \frac{1}{3} = \frac{8}{15} = \frac{1}{60}(31 + 1)$, in agreement with the mass formula (with $|\zeta_K(-1)|/2 = 1/60$ for K). The cuspidal eigenvector is $(3, -5)$.

Out-of-sample match. Independently of the geometry we form the target by brute-force point counting of the elliptic curve $31.1-a1/K$, $[a_1, \dots, a_6] = [1, \varphi + 1, \varphi, \varphi, 0]$, setting $a_q = N(\mathfrak{q}) + 1 - \#E(\mathcal{O}_K/\mathfrak{q})$. No L -function table is read. The Brandt eigenvalues from (1) agree with a_q at all 44 good prime ideals of norm ≤ 200 , both Galois-conjugate ideals resolved individually at every split norm (`data/level31_per_ideal.json`); the level ideal itself is excluded from the good-prime comparison, and its Brandt value -1 is exactly the Steinberg datum consumed by the Atkin–Lehner sign check below.

Verified (computational, reproducible)

Split-prime completeness. For a split rational prime $q = qq'$ the two Hecke eigenvalues $a_q, a_{q'}$ are the values at the totally positive generator ϖ and at its conjugate $\sigma(\varpi)$. The geometry reproduces the *full set* $\{a_q, a_{q'}\}$ — not one representative up to the global Galois involution. For $q = 11, \dots, 89$:

q	11	19	29	41	59	61	71	79	89
$\{a_q, a_{q'}\}$	$\{-4, 4\}$	$\{-4, 4\}$	$\{-2\}$	$\{-6\}$	$\{-4, 12\}$	$\{-2, 6\}$	$\{-8, 0\}$	$\{0, 16\}$	$\{-6, 10\}$

matching the point-counted target exactly (singletons at $q = 29, 41$ occur where $a_q = a_{q'}$).

Atkin–Lehner sign. At $\mathfrak{p}_{31}$ the curve $31.1-a1$ has nonsplit multiplicative reduction ($\#E_{\text{ns}}(\mathbb{F}_{31}) = q + 1$, i.e. $a_{\mathfrak{p}_{31}} = -1$). With the standard newform convention $W_{\mathfrak{p}} = -a_{\mathfrak{p}}$ at a prime of multiplicative reduction exactly dividing the level, the Atkin–Lehner (Fricke) eigenvalue is $W_{31} = -a_{\mathfrak{p}_{31}} = +1$, consistent with the Brandt data.

4 A family, by dimension

The orbit count of §2 gives the Brandt dimension h , and hence the cuspidal dimension $h - 1$ at every prime level with no further computation. (The Eisenstein subspace at prime level is 1-dimensional because $K = \mathbb{Q}(\sqrt{5})$ has narrow class number one, so there is a single Eisenstein class, the all-ones vector.) Table 1 compares the geometric output with Dembélé’s table [7] at every split prime level in his displayed range ($31 \leq N \leq 89$); the smaller split prime norms (11, 19, 29) all give cuspidal dimension 0, consistent with 31 being the smallest level carrying a cusp form.

$N(\mathfrak{p})$	31	41	59	61	71	79	89
geometry (cusp. dim)	1	1	0	2	1	1	1
Dembélé [7]	1	1	0	2	1	1	1
type	ell.	ell.	—	gen. 2	ell.	ell.	ell.

Table 1: Prime-level cuspidal dimensions over $\mathbb{Q}(\sqrt{5})$, geometry vs Dembélé. 7/7. The two informative cases are 59 (no weight- $(2, 2)$ cusp form, only Eisenstein [7, Ex. 3.6]) and 61 (the first form with coefficients outside $\mathbb{Q}$, a genus-2 surface [7, Rem. 4.4]).

The point of Table 1 is that the orbit count is not tuned to one example: with no free parameters it reproduces a non-obvious *sequence*, including a vanishing (59) and a jump to a

non-rational form (61). The same computation continues to larger split levels at no extra cost; we do not tabulate those here, as we have not cross-checked them against an independent source.

5 A family, by eigenvalue: level 41

We repeat §3 verbatim at $\mathfrak{p}_{41}$ ($N = 41$, split): the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{41})$ has two orbits, $\mu = (12, 30)$, $h = 2$, cuspidal dimension 1. The cuspidal eigenvalues are integral, self-adjoint, and satisfy the Ramanujan bound $|a_q| \leq 2\sqrt{N(q)}$ at every prime tested.

For the out-of-sample target we point-count Dembélé's curve $E_{41} : y^2 + \varphi y = x^3 - \varphi x^2$ ([7, Tab. 4], $[a_1, \dots, a_6] = [0, -\varphi, \varphi, 0, 0]$).

Verified (computational, reproducible)

Second form, out-of-sample, no fit. Geometric eigenvalue sets $\{a_q, a_{q'}\}$ vs point-counted E_{41} :

q	11	19	29	31	59	61	71	79	89
geom	$\{-2, 5\}$	$\{-1, 6\}$	$\{2, 9\}$	$\{-10, 4\}$	$\{-3, 4\}$	$\{-8, 6\}$	$\{-12, 9\}$	$\{-11, -4\}$	$\{-8, -1\}$
E_{41}	$\{-2, 5\}$	$\{-1, 6\}$	$\{2, 9\}$	$\{-10, 4\}$	$\{-3, 4\}$	$\{-8, 6\}$	$\{-12, 9\}$	$\{-11, -4\}$	$\{-8, -1\}$

9/9. The same construction, re-pointed to a different level, recovers a different eigenform eigenvalue-for-eigenvalue.

6 The non-rational form: level 61

At $\mathfrak{p}_{61}$ ($N = 61$, split) the A_5 -action on $\mathbb{P}^1(\mathbb{F}_{61})$ has *three* orbits, $\mu = (12, 20, 30)$, $h = 3$, so the cuspidal dimension is 2. By [7, Rem. 4.4], 61 is the smallest level whose newform has coefficients outside $\mathbb{Q}$: the associated Shimura curve has genus 2, and its Jacobian is a modular abelian surface with real multiplication by $\mathbb{Q}(\sqrt{5})$. The $h = 2$ shortcut of §3 does not apply; instead we read the eigenvalue off the 3×3 matrix.

The all-ones vector is the Eisenstein eigenvector (eigenvalue $N(q) + 1$), so the characteristic polynomial of $B(q)$ factors over $\mathbb{Q}$ as $(x - (N(q) + 1))g_q(x)$, where $g_q(x) = x^2 - t_q x + n_q$ is the characteristic polynomial of the 2-dimensional cuspidal block. Comparing traces and determinants,

$$t_q = \text{tr } B(q) - (N(q) + 1), \quad n_q = \frac{\det B(q)}{N(q) + 1} \in \mathbb{Z}, \quad (2)$$

a priori two rational integers. Whether the two cuspidal eigenvalues are rational (two forms) or an irrational conjugate pair (one form with real multiplication) is then decided by the discriminant $t_q^2 - 4n_q$ — which we compute, rather than assume.

Verified (computational, reproducible)

Real multiplication, reconstructed. From (2) the discriminant $t_q^2 - 4n_q$ equals $5 \cdot$ (square) at every prime $N(q) \in \{4, 9, 11, 19, 29, 49\}$. Hence g_q is irreducible over $\mathbb{Q}$ with roots $a_q, \sigma(a_q) \in \mathbb{Z}[\varphi]$ a Galois-conjugate pair: a *single* eigenform with real multiplication by K , not two rational forms. At the inert primes (one prime per rational prime, hence unambiguous) the eigenvalue matches [7] exactly:

$$a_4 = 2\varphi - 2, \quad a_9 = -\varphi - 2, \quad a_{49} = -4\varphi + 2,$$

i.e. $(t_q, n_q) = (-2, -4), (-5, 5), (0, -20)$ respectively. The Ramanujan bound holds on both real embeddings throughout.

The real multiplication is not merely consistent with the data; it follows from it:

Proposition 1. *If g_q has non-square discriminant for even a single q , then the 2-dimensional cuspidal Brandt module at level $\mathfrak{p}_{61}$ is irreducible over $\mathbb{Q}$, the (commutative) Hecke algebra acts on it through the quadratic field $\mathbb{Q}(\sqrt{t_q^2 - 4n_q})$, and every Hecke eigenvalue lies in that field. In particular, $t_4^2 - 4n_4 = 20 = 5 \cdot 2^2$ gives $\mathbb{Q}(\sqrt{5})$, so the level-61 eigenform has real multiplication by $\mathbb{Q}(\sqrt{5})$ and all its eigenvalues lie in $\mathbb{Z}[\varphi]$.*

Proof. On the 2-dimensional cuspidal space, $t_q^2 - 4n_q$ non-square makes g_q irreducible over $\mathbb{Q}$, so T_q acts irreducibly. Any operator commuting with such an irreducible quadratic operator lies in $\mathbb{Q}[T_q] \cong \mathbb{Q}(\sqrt{t_q^2 - 4n_q})$, a field; the Hecke operators all commute and preserve the space, so they lie in $\mathbb{Q}[T_q]$, and every cuspidal Hecke eigenvalue lies in $E := \mathbb{Q}(\sqrt{t_q^2 - 4n_q})$. The value $t_4^2 - 4n_4 = 20$ fixes $E = \mathbb{Q}(\sqrt{5})$; integrality of the Brandt matrices puts the eigenvalues in $\mathcal{O}_K = \mathbb{Z}[\varphi]$. $\square$

So the geometry does not merely *count* the dimension-2 space at 61, nor merely match six published values: it *proves*, from a single discriminant, that the form has real multiplication by $\mathbb{Q}(\sqrt{5})$, and reconstructs its irrational Hecke eigenvalues.

7 The closure certificate algebra

The recurring structural fact above is that each Hecke operator carries a positive invariant form. We isolate this as a small certificate algebra, both because it is the cleanest statement of what the geometry guarantees and because it is the bridge to §8.

Definition 2 (Positive symmetrizer). *A positive symmetrizer for $T \in M_n(\mathbb{R})$ is a symmetric positive definite G with $GT = T^\top G$. Then T is self-adjoint for the inner product $\langle u, v \rangle_G = u^\top Gv$, so it has real spectrum and a G -orthonormal eigenbasis.*

(This is the self-adjoint, “lossless” end of a broader closure taxonomy — dissipative $T^\top GT \prec G$, isometric $T^\top GT = G$ — which we do not need here: the Brandt operators are not contractions, having Eisenstein eigenvalue $N(q) + 1 > 1$.)

Proposition 3. *For every $q \nmid \mathfrak{p}$ the Brandt operator $B(q)$ admits the positive symmetrizer $G = \text{diag}(\mu)$, hence has real spectrum and a μ -orthonormal eigenbasis.*

Proof. The raw incidence count $C_{IJ} = \#\{(x, \gamma) : x \in \text{orbit } I, N(\gamma) = \varpi, x\iota_{\mathfrak{p}}(\gamma) \in \text{orbit } J\}$ satisfies $C_{IJ} = C_{JI}$ via the involution $(x, \gamma) \mapsto (x\gamma, \bar{\gamma})$ (using $\gamma\bar{\gamma} = N(\gamma)$ acting as a scalar on $\mathbb{P}^1$). Since $B(q)_{IJ} = C_{IJ}/(120\mu_I)$ by (1), this gives $\mu_I B(q)_{IJ} = \mu_J B(q)_{JI}$, i.e. $GB(q) = B(q)^\top G$. $\square$

Real spectrum is thus elementary. The sharper *Ramanujan bound* $|a_q| \leq 2\sqrt{N(q)}$ that we observe at every tested prime is *not* elementary: it is the Ramanujan–Petersson bound for these Hilbert cusp forms, transported through Jacquet–Langlands and proved by Eichler–Shimura/Deligne [9, 8]; in this paper we record it as a property satisfied by the computed eigenvalues, not as something we prove. The Brandt operators are, in short, the *positive, finite, real-spectrum* fragment of the arithmetic.

8 Scope: what the finite computation does not reach

Nothing in §3–§7 depends on this section, and it contains no new theorem. We include it only to state precisely what the verified structure does *not* reach.

The rigorous statement. For an L -function, RH is equivalent to genuine positivity criteria: the Weil explicit-formula positivity [14] and Li’s criterion $\lambda_n \geq 0$ for all n [10, 3]. These are theorems (equivalences); they concern the *global, analytic* L -function and an *infinite* family of positivity conditions.

What the geometry delivers, and what it does not.

Verified (computational, reproducible)

The icosian construction delivers the *coefficient side*: the Hecke eigenvalues a_q , equivalently the local factors of the automorphic L -functions, with real spectrum (Prop. 3) and the observed Ramanujan bound. This is finite, per-prime, local data.

Open / not claimed

RH / GRH is the global, analytic statement that the assembled L -function has no off-line zero — the infinite positivity of Weil/Li. A finite Brandt computation produces eigenvalue *coefficients*, not zeros, and supplies none of this positivity. Programs that would close the gap — a self-adjoint operator with the zeros as spectrum (Hilbert–Pólya), or Connes’s arithmetic-site trace formula [4] — remain open; we make no contribution to them. Mirror symmetry (the functional equation) alone does not force zeros onto the line: the Davenport–Heilbronn phenomenon [6] exhibits Dirichlet series with the functional equation yet off-line zeros.

9 Reproducibility

All results are produced by pure-Python scripts in the accompanying bundle (`route_b/`); none requires a computer-algebra modular package. The anti-circularity discipline is enforced in code: targets are computed by `point_count_target.py` (brute-force $\#E(\mathcal{O}_K/q)$) and read only at comparison time.

script	result
<code>ideal_classes.py</code>	level-31 orbits, mass 8/15, $h = 2$
<code>brandt_matrices.py</code>	Brandt/Hecke operators (1), invariants
<code>brandt_level.py</code>	the engine at an arbitrary split prime level
<code>hecke_compare.py</code>	level 31, original out-of-sample protocol
<code>level31_full_ideals.py</code>	level 31, all 44 ideals $N \leq 200 \rightarrow$ <code>data/level31_per_ideal.json</code>
<code>second_form_level41.py</code>	level 41, 9/9 vs $E_{41} \rightarrow$ <code>data/level41_results.json</code>
<code>genus2_form_level61.py</code>	level 61 genus-2, RM eigenvalues $\rightarrow$ <code>data/level61_results.json</code>
<code>multilevel_dimensions.py</code>	dimension sequence (Table 1) $\rightarrow$ <code>data/dimension_sequence.json</code>
<code>no_fit_guard.py</code>	anti-circularity check over the full geometry path

10 Conclusion

We have shown, by an independent geometric route, that the 600-cell / icosian order reproduces a *family* of Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$: a non-obvious dimension sequence (7/7), the full Hecke systems of two rational eigenforms (out-of-sample, no fit), and the irrational eigenvalues of the genus-2 form at level 61. We have framed the self-adjoint Hecke operators in a closure/resonance certificate algebra and used it to state exactly what separates this verified, finite, coefficient-side structure from the Riemann Hypothesis: the global analytic positivity, the lossless self-adjoint engine, which we characterize and do not construct. The value of the work is a clean, reproducible realization of known arithmetic from polytope geometry, and an honest account of the gap that remains.

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